FINITELY MANY POINTS OF A QUASI-AFFINE SCHEME OR A SCHEME WITH AN AMPLE INVERTIBLE SHEAF LIE IN AN AFFINE OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction of the four cases to the projective case, the covering of the finite set by finitely many standard opens avoiding the complement, and the graded argument merging them into a single standard open whose intersection with X is affine.

Editorial note. The label lemma-ample-finite-set-in-affine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source properties.tex –label lemma-ample-finite-set-in-affine –start 5460 –end 5507 –local-id 28036 returned RESULT=OK (L28036 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-properties/candidates-halfA.json.

Editorial note. The proof contains the sentence 'We urge the reader to prove case (2) directly for themselves.' — a reading suggestion for one of the four hypotheses, while the proof itself reduces that case and the others to the projective case, so the statement is counted and the sentence is recorded here verbatim.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of quasi-affine schemes and of schemes with an ample invertible sheaf. Same chapter, the definition of a quasi-affine scheme, the identification of an affine scheme with a projective spectrum of a polynomial ring, and the proposition characterizing schemes with an ample invertible sheaf as open subschemes of projective spectra. Supplies the reduction of cases (1), (2) and (3) to case (4).
- Standard opens form a basis of the projective spectrum. Algebra, the description of the standard opens D_+(f) for homogeneous elements as a basis of the topology of the projective spectrum. Supplies the covering of the finite set by finitely many standard opens avoiding the closed complement.
- Existence of a homogeneous element avoiding finitely many primes. Algebra, the lemma producing a homogeneous element of a graded ideal which avoids finitely many primes not containing the ideal. Supplies the single

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homogeneous element f whose standard open contains the finite set and avoids the complement.

Source reference. Original lemma 00JS: sources/stacks-chow/tags.txt, source lines 726–726. Prerequisite pointer invoked at proof line(s) 5501; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01P6: sources/stacks-properties/pinned-original-source.tex, source lines 2624–2624. Prerequisite pointer invoked at proof line(s) 5476; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01Q3: sources/stacks-properties/pinned-original-source.tex, source lines 4774–4774. Prerequisite pointer invoked at proof line(s) 5480; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. Assume either*

1. *The scheme X is quasi-affine.*
2. *The scheme X is isomorphic to a locally closed subscheme of an affine scheme.*
3. *There exists an ample invertible sheaf on X.*
4. *The scheme X is isomorphic to a locally closed subscheme of Proj(S) for some graded ring S.*

Then for any finite subset $E \subset X$ there exists an affine open $U \subset X$ with $E \subset U$.

Proof. By Properties, Definition 01P6 a quasi-affine scheme is a quasi-compact open subscheme of an affine scheme. Any affine scheme $\mathrm{Spec}(R)$ is isomorphic to $\mathrm{Proj}(R[X])$ where $R[X]$ is graded by setting $\deg(X) = 1$. By Proposition 01Q3 if X has an ample invertible sheaf then X is isomorphic to an open subscheme of $\mathrm{Proj}(S)$ for some graded ring S. Hence, it suffices to prove the lemma in case (4). (We urge the reader to prove case (2) directly for themselves.)

Thus assume $X \subset \mathrm{Proj}(S)$ is a locally closed subscheme where S is some graded ring. Let $T = \overline{X} \setminus X$. Recall that the standard opens $D_+(f)$ form a basis of the topology on $\mathrm{Proj}(S)$. Since E is finite we may choose finitely many homogeneous elements $f_i \in S_+$ such that

$$E \subset D_+(f_1) \cup \ldots \cup D_+(f_n) \subset \mathrm{Proj}(S) \setminus T$$

Suppose that $E = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_m\}$ as a subset of $\mathrm{Proj}(S)$. Consider the ideal $I = (f_1, \ldots, f_n) \subset S$. Since $I \not\subset \mathfrak{p}_j$ for all $j = 1, \ldots, m$ we see from Algebra, Lemma 00JS that there exists a homogeneous element $f \in I$, $f \notin \mathfrak{p}_j$ for all $j = 1, \ldots, m$. Then $E \subset D_+(f) \subset D_+(f_1) \cup \ldots \cup D_+(f_n)$. Since $D_+(f)$ does not meet T we see that $X \cap D_+(f)$ is a closed subscheme of the affine scheme $D_+(f)$, hence is an affine open of X as desired. □